Course Overview

MGMT 675: AI-Assisted Financial Analysis

RICE | BUSINESS
Jones Graduate School of Business

Meet your Prof

- At Rice since 2009, in Jones and in Econ Dept.
  - teach core finance, quantitative investments, investments theory, and python for business research
  - to PhD and Masters in Data Science students
- Previously at Northwestern, Indiana, Washington Univ. in St. Louis, and Texas A&M. Associate Dean at Wash U.
- Former and current editor and associate editor of several journals. Two textbooks (derivatives and PhD asset pricing theory).
- Course materials and info at kerryback.com

Course Overview

Motivation for this course

- How will AI affect finance jobs?
- Basic template for finance chatbot
  - Python code for financial analysis
  - App that encapsulates python code
  - App makes API calls to LLMs to process user's (natural language) input and pass structured version to python code
- Random web cites: 40+% of all new code written by AI, 80+% of developers use AI assistance

Learning objectives

- Learn to use AI to write python code to perform financial analyses
- Get additional practice in financial analyses
- Obtain a basic understanding of
  - Python
  - Making API calls to LLMs
  - Building simple apps

Platforms

- Julius.ai to write and execute python code
- Google Colab as alternative python environment (free)
- VS Code (free) and Cursor as alternative local environments
- Streamlit Cloud for hosting apps (free)
- OpenAI for API calls
- HuggingFace as alternative for API calls (free)

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Assignments

- 6 group assignments, due by midnight on Thursdays beginning this week
- Form groups on canvas of no more than six students
- We'll break out into groups on most days
- Each assignment is to provide a link to a Julius workflow that will accomplish a specific financial analysis

Julius

Get a Julius account

- Julius.ai provides a 50% academic discount. Sign up for a free account, then send an email using your Rice email account to team@julius.ai and ask for the academic discount. They will respond with a promo code to use.
- The Lite account ($8 per month after discount) allows 250 messages per month and may be ok.
- Standard account ($18 per month after discount) allows unlimited messages.

Warm-Up

Click "+ New" to start a chat (thread)with Julius. Ask it to do the following (you could start a new chat for each topic):

- Ask Julius to use yfinance 0.2.54 to get AAPL's closing price from Yahoo Finance, plot it, and save as a jpeg.
- Ask Julius to use pandas datareader to get the 10-year Treasury yield from FRED, plot it, and save as a jpeg.
- Ask Julius to plot the payoff diagram of a call option with a strike of 100.
- Ask Julius what data it needs to compute the Black-Scholes value of a call option. Supply the data and ask for the value.

Julius Workflows

- Scroll down the "Workspace" menu to find and select "Workflows."
- Move to the "Sandbox" tab and select "New Sandbox."
- The "Prompt" menu on the right includes a "User Prompt" option and a "Prompt" (tell Julius to run an analysis) option.
- Each workflow should start with a user prompt, asking the user to supply data and maybe asking the user what output the user wants.
- Specify in a Julius prompt the steps Julius should take with the information provided by the user.
- Follow up with another user prompt if needed, etc.

Example Workflow

Create a workflow including the following:

- User prompt: Ask the user for a ticker and a start date.
- Julius prompt: Use yfinance to get the closing price of the stock from Yahoo Finance beginning at the start date. Plot the price. Output the plot to a jpeg file and show it to the user.

Excel, Python, and AI

Excel and Python

- Data in Excel $\rightarrow$ python is easy
- Python output $\rightarrow$ Excel is easy
- Formulas in Excel $\rightarrow$ python is possible via AI
- Python formulas $\rightarrow$ Excel is possible via Python in Excel

Excel $\rightarrow$ Python via AI

- Default is to assume each worksheet is a single table and read the numbers (not the formulas) into a dataframe.
- Can ask Julius to read the formulas.
- Can also warn Julius that there are multiple tables in a worksheet if that is the case.
- Python implements formulas as functions. Can ask Julius to read formulas and convert to functions. For example, a function can compute a row of a cash flow table from other rows.

Building models in python with AI

- Read and mimic Excel model, or
- Ask AI to build a model from scratch
- Both are often iterative processes. Need to correct AI mistakes.

Saving and reusing models

- As Julius workflow: Copy working code and paste it into a "Prompt" cell when creating a workflow. Ask Julius to use the code (works somewhat).
- Copy code and paste into a Jupyter notebook on your hard drive or Google Drive (for Colab).
- Create a chatbot app that runs the code.